16-12-2025

Lab - 4

Simple Linear Regression

Step 1. Import the necessary libraries

In [ ]:
```
import pandas as pd
```

In [ ]:
```
from sklearn.model_selection import train_test_split
```

Step 2. Import the dataset

In [ ]:
```
df = pd.read_csv("50_Startups.csv",sep=",")
```

In [ ]:
```
df.head()
```

Step 3 . Check the State Column

In [ ]:
```
df['State'].value_counts()
```

Step 4 . Splitting dataset in to input and output

In [ ]:
```
x = df.drop('Profit',axis=1)
y = df['Profit']
```

In [ ]:
```
x.head()
```

In [ ]:
```
y.head()
```

Step 5 . Convert state Column into Numeric Column

Step 5.1 . Perform Transformation

In [ ]:
```
x1 = pd.get_dummies(x,columns=['State'],drop_first=True)
```

In [ ]:
```
x1.head()
```

Step 6 . Dummy variable trap

In [ ]:
```
# Already Performed using | drop_first =True
```

Step 7 Splitting dataset in to Train and Test

In [ ]:
```
x_train, x_test, y_train, y_test = train_test_split(x1,y,random_state=42,test_size=
```

Step 8 Import LinearRegression model from linear_model family

In [ ]:
```
from sklearn.linear_model import LinearRegression
```

In [ ]:
```
model = LinearRegression()
```

Step 9 Fit the data

In [ ]:
```
model.fit(x_train,y_train)
```

Step 10 Predict the data

In [ ]:
```
y_predict = model.predict(x_test)
```

Step 11 Display Result

In [ ]:
```
# y_test and y_predict
import matplotlib.pyplot as plt
```

In [ ]:
```
plt.scatter(x_test['Administration'],y_test)
plt.scatter(x_test['Administration'],y_predict)
```

RSS

In [ ]:
```
import numpy as np
```

In [ ]:
```
np.sum((y_test.values - y_predict) ** 2)
```

In [ ]:
```
len(y_test)
```

In [ ]:
```
from sklearn.metrics import mean_squared_error
```

In [ ]:
```
mean_squared_error(y_test.values,y_predict) * len(y_predict)
```

R Square

In [ ]:
```
from sklearn.metrics import r2_score
```

In [ ]:
```
r2_score(y_test,y_predict)
```

Now use Polynomial Regression on Position_Salaries dataset

In [ ]:
```
df1 = pd.read_csv('Position_Salaries.csv',sep=',')
```

In [ ]:
```
df1
```

In [ ]:
```
plt.scatter(df1['Level'] , df1['Salary'])
```

In [ ]:
```
a = df1.iloc[::,1:2:]
b = df1['Salary']
```

In [ ]:
```
a
```

In [ ]:
```
a_train,a_test,b_train,b_test = train_test_split(a,b,random_state=42,test_size=0.2)
```

In [ ]:
```
from sklearn.preprocessing import PolynomialFeatures
```

In [ ]:
```
poly = PolynomialFeatures(degree=2)
```

In [ ]:
```
x2 = poly.fit_transform(a_train)
```

In [ ]:
```
x2
```

In [ ]:
```
model1 = LinearRegression()
```

In [ ]:
```
model1.fit(x2,b_train)
```

In [ ]:
```
b_poly_predict = model1.predict(poly.fit_transform(a_test))
```

In [ ]:
```
b_poly_predict
```